

Testing with Spring

Unit, context, integration and web testing in the Spring ecosystem.

Strategy

Combine fast unit tests with context/integration tests only where they add confidence. Avoid starting the whole application when an isolated unit is enough.

JUnit and Mockito

JUnit provides execution/assertions; Mockito supports isolation. Spring adds context and testing infrastructure.

Web and APIs

For web applications, validate controllers and endpoints at the appropriate level and complement with external API tests when needed.

Data and integrations

Control data, state, transactions and dependencies. Choose consciously between mocks, stubs, containers or real environments.

QA and quality

QA can contribute to strategy, identify integration risk and keep the pyramid from depending only on slow E2E tests.

Installation and configuration - overview

Always use the official website and documentation. Check prerequisites and compatible versions, install for your operating system, configure environment variables or PATH only when required, validate the installation with the recommended command or screen, and run a first project/test before integrating it into the QA workflow.

QA tip

Always connect the practice to product risk and the evidence needed for quality decisions.

Official resources

Site oficial: <https://docs.spring.io/spring-framework/reference/testing.html>

Download / instalação: <https://start.spring.io/>

Documentação: <https://docs.spring.io/spring-framework/reference/testing.html>